Abhinav Girish

ms24btech11003@iith.ac.in | abnv100@gmail.com | +91 6363501636 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.81
Sri Chaitanya College of Education	Class XII (CBSE)	2024	95.80%
Delhi Public School, Bangalore South	Class X (CBSE)	2022	96.40%

WORK EXPERIENCE

AI Engineer Intern, Mercedes-Benz Research & Development India (MBRDI)

Self-Hosted LLM Inference Platform · *vLLM, Qwen2.5-Coder-14B, Nginx, RAG, RAGAS*

- **Architected** and **deployed** an **end-to-end** self-hosted platform on an NVIDIA A6000, serving a code-assistant model to an internal engineering team through Open WebUI and Continue.dev IDE integration over a **TLS-secured endpoint**.
- Improved **reliability and scalability** of the serving layer with **Nginx per-IP rate limiting**, load-testing **throughput scaling 5.4x (35 to 193 tok/s)** from serial to 16 concurrent requests at **zero failures.**
- Designed a **retrieval-augmented generation pipeline** (web/GitHub/markdown ingestion, Chroma) that **improved answer quality 63%** (LLM-judge 2.5 to 4.1/5), measured on a **custom evaluation harness** scoring generation and retrieval quality (**RAGAS: 0.87 faithfulness**) with **automated regression comparison**.

Multi-Agent Orchestration System · *Python, FastAPI, LangGraph, A2A Protocol, AWS Bedrock, MCP*

- Built a **FastAPI backend service** that routes requests via **dynamic capability discovery**, combining skill-tag heuristics with LLM-based fallback routing and **SQLite-backed task persistence**.
- Developed a LangGraph ReAct agent that **integrates** a Black Duck MCP server through **10 tool wrappers**, cutting security-compliance lookups from **12-18 UI clicks to 2-4 chat messages** by exposing vulnerability, license, and policy-violation data via natural language, backed by Claude on AWS Bedrock.

PROJECTS

SpectraBand - hyperspectral band-selection study · *Python, PyTorch, scikit-learn* · GitHub

- Built a **1D CNN** for **hyperspectral pixel classification** on the **Indian Pines** benchmark (**200 bands**) under a **spatially disjoint split** eliminating neighbour leakage, characterizing the **accuracy-vs-band-count crossover** across **90 training runs** -- **uniform subsampling at 10 bands** (**OA 0.712**) beats the **full 200-band** model (**OA 0.650**) by **6.2 points** on **5% of the spectrum**.
- Disproved the **redundancy hypothesis** for that crossover with a **redundancy-penalized mRMR ranking** that closed the gap at only **3 of 9 band counts**, establishing that **greedy univariate selection** fails beyond inter-band correlation alone.

SeekTree - B+ tree index & query optimizer · *Python, SQLite* · GitHub

- Engineered a **B+ tree index** with **node splitting/cascading** and **linked-leaf range queries** from scratch, reducing exact-match lookups from **O(n)** full scan to **O(log n)** index seek -- **62.7ms to 0.002ms** on a 1M-row table (**~26,900x**)
- Built a **cost-based optimizer** selecting **index-seek vs. full-scan** by estimated **selectivity** -- validated against a **SQLite oracle** across **30+ randomized queries** -- and benchmarked the crossover across three **fetch-cost models**, showing the tradeoff is governed by **row-fetch I/O cost** rather than a fixed threshold

DagQueue - DAG-based task scheduler · *Python, FastAPI, SQLite, concurrent.futures* · GitHub

- Engineered a **polling scheduler** that resolves task dependencies and dispatches independent tasks concurrently on a **thread pool**, achieving **~7x throughput** over serial execution on I/O-bound workloads (8 tasks: **8.1s to 1.2s**).
- Built **failure isolation** that propagates a skipped state to downstream dependents of any failed task, preventing indefinite scheduler hangs and guaranteeing **deterministic termination**.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, concurrency & parallelism, REST API design, time & space complexity analysis, Database Management Systems

Backend & Infrastructure: FastAPI, Nginx (reverse proxy, TLS, rate limiting), SQLite, AWS Bedrock, Linux, Git/GitHub

AI/ML & LLM Systems: PyTorch, CNNs, scikit-learn, NumPy, LLM serving & inference (vLLM), Retrieval-Augmented Generation (RAG), LLM evaluation (LLM-as-judge, RAGAS), multi-agent orchestration, Model Context Protocol (MCP)

RELEVANT COURSEWORK

Computer Science: Data Structures and Algorithms, Database Management Systems, Fundamentals of Scientific Computing, Computational Methods

Mathematics: Probability and Random Variables, Linear Algebra, Calculus I & II, Differential Equations, Mathematics for Physics

POSITIONS OF RESPONSIBILITY

Head, Public Relations & Networking -- Entrepreneurship Cell, IIT Hyderabad

- Led a team of 30+ members driving national outreach; organised Founders' Hive across 2 cities (30+ founders, 300+ participants) and coordinated sponsors for E-Summit 2025 & 2026 (12L+ budget).

Internship Coordinator - Office of Career Services, IIT Hyderabad

- Build and manage the internship company pipeline, driving company outreach and coordinating end-to-end drives (JDs, tests, interviews) while bridging student and company requirements.

Core, Corporate Networking & Growth -- Tinkerers' Lab, IIT Hyderabad

- Organised THRUST 2025 for 6,000+ attendees and helped secure a 15L+ sponsorship portfolio.